Author Response to Reviewer Comments

Response to Reviewer 1

Comment 1: Authors evaluated the diagnostic yield according to many clinical and operational outcomes; this approach is quite similar to those used by previous works. I believe that, considering the recent onset of molecular analysis for personalized approach to lung cancer, they should have included also the appropriateness of samplings in providing enough tissue for molecular tumor assessment. Similarly, outcoming the goodness of the sample for semisolid, solid or GG lesion's profile would have been appropriate.

Response 1: Thank you very much and we wish to express our appreciation for your in-depth comments, suggestions, and corrections, which have greatly improved the manuscript. To the best of our knowledge, this is the first systematic review with meta-analyses of randomized controlled trials (RCTs) published to date to investigate the diagnostic yield and total examination time of VBNA and NVBNA bronchoscopy for the diagnosing peripheral pulmonary lesions. As this is the meta-analysis of published randomized controlled trails up-to-date, the included studies in this meta-analysis lacked data on molecular analysis, semisolid, solid or ground glass lesions. So we just evaluated the diagnostic yield according to many clinical and operational outcomes only. We plan to write meta-analysis to address the concerns you have raised in future with availability of the data from RCTs on molecular analysis, semisolid, and solid or ground glass lesions.

Comment 2: Authors found that VBNA provides a statistically significant shortening of the total examination time. I believe it is important to remark that this finding is far from being clinically relevant, I suggest authors to clarify this in the discussion section.

Response 2: Thank you very much for your comment. We have included this in discussion section as “Though VBNA demonstrated a statistically significant shortening of the total examination time, this finding is far from being clinically relevant (mean difference being about 4 minutes), future well designed multi-center RCTs are needed to verify our finding. However, in terms of patients comfort, decreasing overall examination time by 4 minutes, is significant especially in patients undergoing the procedure under local anesthesia.” (Note that this section is now in line 274 of page 13 in revised manuscript)

Comment 3: Considering that it is relevant, I would move the figure S1 from supplementals to main manuscript and would comment it more in deep.

Response 3: We moved Figure S1 from supplemental file to main manuscript as Figure 4 and commented in discussion section as "In subgroup analysis, diagnostic yield of lesions ≤20 mm was higher in VBNA group than in NVBNA group. This was in line with finding of previous meta-analysis.[26] At the same time, Kato et al.[29] demonstrated that the diagnostic yield of small PPL <20 mm in diameter was significantly higher in VBN assisted group than non-VBN assisted group. The use VBNA might improve bronchial path selection more accurately and quickly for small lesion, in contrast to larger lesions which could have several routes reaching to the target lesion. Additionally, a recent study that compared the diagnostic yield of VBN-guided and unguided ultrathin bronchoscopy found that the diagnostic yield was slightly higher for PPLs ≤20 mm in the VBN- ultrathin arm, but the difference was not statistically significant ($p = 0.069$).[30] Although diagnostic yield do not depend solely on the lesion size but it is also affected by the target disease, location and presence or absence of an involved bronchus.[2]" (Note that this section is now in line 279 of page 13 in revised manuscript)

Comment 4: Considering the conclusions of this metanalysis, I believe that authors should discuss the many issues in favour and/or against the use of virtual supported approaches vs other non-virtual assisted ones, i.e. economic burden, availability, etc.

Response 4: We have included many issues in favor of VBN and other non-virtual assisted approaches in conclusion section as "VBN is a form of novel guided bronchoscopy that requires no specific training, and has a low complication rate. It can shorten the positioning time and it is a safe and effective promising technique for pulmonary nodules. VBN improves the diagnostic yield when combined with other methods, such as EBUS or R-EBUS, EBUS-GS-TBLB for PPLs. This technique also helps to abandon the X-ray guidance, thus avoiding significant cumulative radiation dose for the patient and the operator. The detailed analysis of current several bronchoscopic technologies, including advantages and disadvantages is included in Table S1 (Supplementary information)." (Note that this section is now in line 313 of page 14 in revised manuscript)

Comment 5: Still in the discussion, I would raise the concern related to the "expertise bias" that characterizes many studies evaluating diagnostic yield in bronchial endoscopy and whose diagnostic accuracy probably reflects the skillfulness of the interventional pulmunologists carrying out the study. Their extremely high diagnostic yield is unlikely to be generalizable to the real world; this is particularly true for fluoroscopy or radial

EBUS, guide sheath guided sampling. This potential overestimation might affect the results of the metanalysis.

Response 5: Thank you very much for your valuable suggestion. We have included expertise bias in limitation of this study as “Third, the high variability in diagnostic yield between individual trials included in this meta-analysis may be attributed to several factors including expertise of interventional pulmonologist, presence or absence of rapid on-site evaluation (ROSE), and biopsy tools selection. This potential overestimation of diagnostic yield due to expertise bias may affect the meta-analysis results.” (Note that this section is now in line 299 of page 14 in revised manuscript)

Comment 6: not considering some studies because "outlier" is an uncorrect approach that may provide some misleading message. If the authors decided to include this paper in their analysis, no exceptions should be made when interpreting their results. I would change this section according.

Response 6: Thank you very much for your comment. We did not exclude any study from analysis because of outlier. We just showed that after excluding outlier studies there was no significant heterogeneity among studies. So the heterogeneity was mainly due to out outlier studies. In final analysis we included all studies. So there was no exceptions when interpreting our results.

Minor issues:

Comment 1: use the term Systematic instead of Systemic in both the title and the text

Response 1: Thank you for your valuable suggestion. We have used term Systematic instead of “Systemic” in both the title and the text.

Response to Reviewer 2

Comments to the Author

In this meta-analysis, the author compared RCT of virtual bronchoscopic navigation (VBN) versus Non-VBN in the diagnosis of a peripheral pulmonary nodule. Here was no significant difference in the overall yield. There was a slight improvement in the yield when the PPL is less than 2 cm in diameter. The study is interesting to read and well done. I would recommend accepting with minor revisions as below:

Response: Thank you very much and we wish to express our appreciation for your in-depth comments, suggestions, and corrections, which have greatly improved the manuscript.

Comment 1- Page 4- Line8: Definition of traditional PPL is inaccurate- please revise

Response 1- We have revised definition of traditional PPL. (Note that this section is now in line 71 of page 4 in revised manuscript)

Comment 2- Page 4- line 15: the statement is not needed

Response 2- We have modified the statement**.** (Note that this section is now in line 75 of page 4 in revised manuscript)

Comment 3- Page 4- line41: reference 11 is an inaccurate reference for that statement- reference 11 study did not include VNB- please place appropriate reference.

Response 3- We have replaced reference 11 with appropriate reference.

"11. Kemp SV. Navigation Bronchoscopy. Respiration 2020; 99: 277–286."

Comment 4- Page4- Line 51: reference 12 did not show better yield for VBN- please remove that reference

Response 4- We have replaced reference with more appropriate reference.

"12. Tachihara M, Tamura D, Kiriu T, et al. Bronchoscopy Using Virtual Navigation and Endobronchial Ultrasonography with a Guide Sheath (EBUS-GS) with or without Fluoroscopy for Peripheral Pulmonary Lesions. Kobe J Med Sci 2018; 63: E99–E104."

Comment 5- Page 5- line 23: remove "lung lobe"

Response 5- We have removed "lung lobe". (Note that this section is now in line 102 of page 5 in revised manuscript)

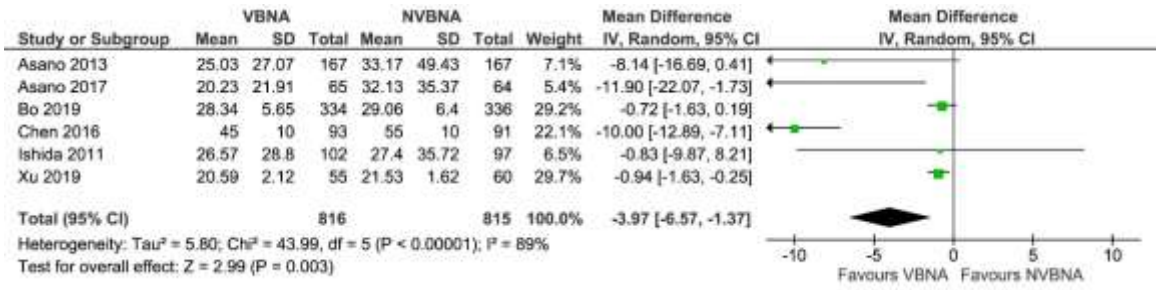
Comment 6- Table 1: add the number of participants in each study, average lesion size and if radial EBUS was used or not; You may remove the age and the gender as it does not add any information

Response 6- We have added the number of participants in each study and total examination in Table 1. We have added Figure 4 in main manuscript and average size of lesion is given in forest plot.

Radial EBUS was used in study by Bo et al and we have included in discussion section as "The choice of bronchoscopic modalities such as CT-guided biopsy and VBN and/or r-EBUS or conventional bronchoscopy varies from patient to patient. EBUS requires operation expertise and enables direct visualization target lesion. In addition to the VBN, study by Bo et al.[17] used r- EBUS for the diagnosis of peripheral pulmonary lesion. Of note, the diagnostic yields between combined group (VBN + EBUS) and EBUS group were similar. The absence of benefit seen with VBN and r-EBUS may have been due to patient selection, with more difficult cases being selected for VBN and r-EBUS." (Note that this section is now in line 262 of page 12 in revised manuscript)

Comment 7- Figure3: reads as favor NVBNA in term of examination time, which is opposite to what was described in the text – please adjust

Response 7- Thank you very much for your correction. There was mistake in labelling of Figure 3, we have corrected it.



Comment 8- please add a discussion about the effect of the radial EBUS in the included studies, ROSE, in the reported studies, was any other form of navigation used?

Response 8: Radial EBUS was used in study by Bo et al. and we have included in discussion section as "The choice of bronchoscopic modalities such as CT-guided biopsy and VBN and/or r-EBUS or conventional bronchoscopy varies from patient to patient. EBUS requires operation expertise and enables direct visualization target lesion. In addition to the VBN, study by Bo et al.[17] used r- EBUS for the diagnosis of peripheral pulmonary lesion. Of note, the diagnostic yields between combined group (VBN + EBUS) and EBUS group were similar. The absence of benefit seen with VBN and r-EBUS may have been due to patient selection, with more difficult cases being selected for VBN and r-EBUS." (Note that this section is now in line 262 of page 12 in revised manuscript)

Furthermore, we have included in limitation section also about the effect of the radial EBUS and ROSE as "Third, the high variability in diagnostic yield between individual trials included in this meta-analysis may be attributed to several factors including expertise of interventional pulmonologist, presence or absence of rapid on-site evaluation (ROSE), and biopsy tools selection. This potential overestimation of diagnostic yield due to expertise bias may affect the meta-analysis results." (Note that this section is now in line 299 of page 14 in revised manuscript)

Response to Reviewer 3

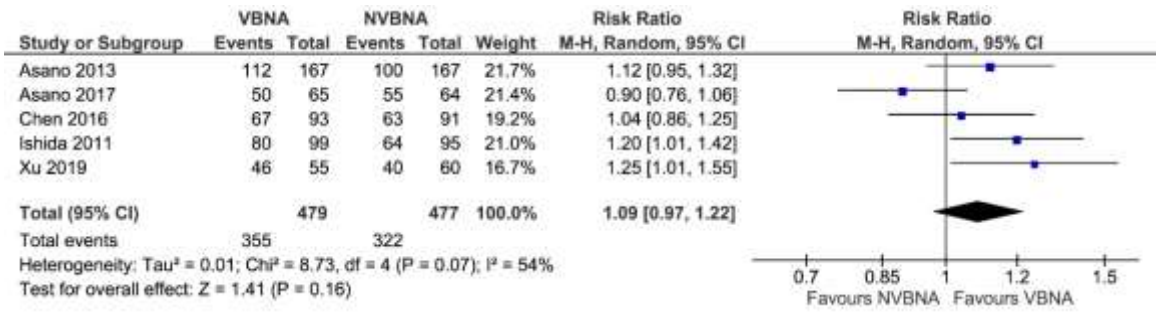
Comments to the Author

This is a good paper to look at what we can find from meta-analysis now. I have one major comment about a study included in the analysis and a few other minor notes.

Comment 1: The inclusion of the study by Bo that used EBUS seems to problematize the results in a similar manner to how the Jiang meta-analysis was affected by including ENB. By using the study by Bo you're including biopsies performed with ultrasound image guidance in with the "control" NVBNA, which in the other studies has no comparable guidance. Even if all I knew about the Bo study was that they compared EBUS-GS with EBUS-GS & VBN, I would expect much less of a difference in diagnostic yield than you'd see with NVBNA compared to VBNA. That's how the data plays out in your analysis. The study by Bo is by far the most weighted in your analysis of overall diagnostic yield and has the 2nd worst M-H of all the studies. If it was omitted it appears looking at your Forest plot that you would likely find that VBNA does improve overall diagnostic yield. I think it makes sense to not include it because the data from it that you're including in the NVBNA group has a different form of image guidance (EBUS) in it as opposed to all the rest of the data. At a minimum I think you should show how things would look if you didn't lump EBUS in with a subset of your data. The addition of EBUS should have some impact on most of the outcomes you're looking at.

Response 1: Thank you very much and we wish to express our appreciation for your in-depth comments, suggestions, and corrections, which have greatly improved the manuscript. We have included all the studies in this meta-analysis with strict inclusion and exclusion criteria's. In addition to the VBN, study by Bo et al.[17] used r- EBUS for the diagnosis of peripheral pulmonary lesion. Of note, the diagnostic yields between combined group (VBN + EBUS) and EBUS group were similar. Even after omitting study by Bo et al. VBNA does improve overall diagnostic yield. That's why we think that Study by Bo et al does not have impact on the outcomes we evaluated.

Below is forest plot after removing study by Bo et al.



As for more minor notes:

Comment 1: On page 9 line 12 (last sentence before the "Secondary Outcomes" you say "there was no heterogeneity among studies" and I assume you meant "no significant heterogeneity"

Response 1: Thank you very much! We have corrected it as “"there was no significant heterogeneity among studies." (Note that this section is now in line 89 of page 9 in revised manuscript)

Comment 2: In Figure 3, I know it is just following standard methods for how to show a Forest plot, but when I look at it and read under the x-axis "Favours NVBNA" and "Favours VBNA" and see the data on the side that says "Favours NVBNA" it initially makes me think that NVBNA does better on examination time.

Response 2: Thank you very much for your correction. There was mistake in labelling of Figure 3, we have corrected it.

